Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH (BIOTECHNOLOGY)

END SEMESTER EXAMINATIONS, JAN 2025 (MAKE-UP)

SUBJECT: BIOREACTION ENGINEERING (BIO 3104)

REVISED CREDIT SYSTEM

Time: 3 Hours **(01.01.2025)** MAX. MARKS: 50

Instructions to Candidates:

- Answer **ANY FIVE FULL** the questions.
- Missing data may be suitable assumed.

	QUESTION	M
1A.	A particular product is being produced using *Pseudomonas* species in a batch culture with following stoichiometry. Come up with a mechanism that is consistent with the experimentally found rate equation for the following reaction $2A + B \rightarrow A_2B$ Experimental rate: $+r_{A2B} = k [A] [B]$	**4**
1B.	Explain: Order of the reaction, Molecularity, Rate constant	**3**
1C	Find the reaction rate in the following cases: 1. Homogeneous reaction system of hydrolysis of sucrose in a shake flask with 200 ml reaction volume and 9000 micromoles product formed in 30 minutes 2. Heterogeneous reaction system of hydrolysis of sucrose in an immobilized packed bed bioreactor with 100 g beads and 6000 micromoles product formed in 30 minutes.	**3**
2A.	At room temperature sucrose is hydrolyzed by the catalytic action of the enzyme invertase as follows: $sucrose + water \xrightarrow{Invertase} Fructose + Glucose$ Starting with a sucrose concentration C_{A0} = 1.0 millimol/liter and an enzyme concentration C_{E0} = 0.01 millimol/liter, the following kinetic data are obtained in a batch fermenter CA, mM: 0.84, 0.68, 0.53, 0.38, 0.27, 0.16, 0.09, 0.04, 0.018, 0.006, 0.0025 Time, h: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 Determine whether these data can be reasonably fitted by a kinetic equation of the Michaelis-Menten type, or	**4**

	$-r_A = \frac{k_3 C_A C_{E0}}{C_A + C_M}$ where C_M = Michaelis constant If the fit is reasonable, evaluate the constants k_3 and C_M . Solve by the integral method.	
2B.	For elementary reactions in series A → R →S, K1=K2, at t=0, CA=CA0 and CR0=CS0=0 Find the maximum concentration of R and when it is reached	**4**
2C	Substrates A (C_{A0}=10M) and B (C_{B0}=20M) react by 2nd order kinetics, non-elementary reaction type: A + 2B → Product in a batch reactor. 50% of A is converted in 12 hours. How much longer would it take to reach 75% conversion?	**2**
3A.	A zero order homogeneous gas reaction A → γ R proceeds in a constant-volume batch reactor, π=1 when t=0 and π=1.5 when t=1 hour. If the same reaction, same feed composition and initial pressure proceeds in a constant pressure setup, find V at t=1 if V=1 at t=0.	**4**
3B.	Pure gaseous A at about 3 atm and 30ºC (120 mmol/liter) is fed into a 1-liter mixed flow reactor at various flow rates. There it decomposes, and the exit concentration of A is measured for each flow rate. From the following data find a rate equation to represent the kinetics of the decomposition of A. Assume that reactant A alone affects the rate: A → 3 R	**4**
3C	Differentiate between Ideal Plug flow and Ideal Mixed Flow Reactors	**2**
4A.	Explain how to find the best reactor set up when two unequal size MFR's are connected in series. Use the graphical approach.	**3**
4B.	For an irreversible first-order liquid-phase reaction (CA0 = 10 mol/liter) conversion is 90% in a plug flow reactor. If two-thirds of the stream leaving the reactor is recycled to the reactor entrance, and if the throughput to the whole reactor-recycle system is kept unchanged, what does this do to the concentration of reactant leaving the system?	**3**
4C.	At present the elementary liquid reaction A + B →R + S takes place in a plug flow reactor using equimolar quantities of A and B. Final desired conversion is 96 % , C_{A0}=C_{B0}=1 mol/liter. If a mixed reactor ten times as larger as plug flow reactor were hooked up in series with the existing unit, which unit should come first and by what fraction could production be increased for that setup?	**4**
5A.	Explain the operating diagram of chemostat with cell cultures. How do you operate the chemostat at steady state-Explain?	**4**
5B.	Explain in the various accessories needed for the operation of batch fermenter with cell cultures. Derive the expression for the batch reaction time for the Growth of biomass from X_0 to X_f during batch cultivation of bakers yeast	**3**
5C	What do you mean by variable volume batch reactor. Explain any one variable volume batch reactor used industrially. Write the rate expression for variable volume system.	**3**

υ_0, liter/min	0.06	0.48	1.5	8.1
C_A, mmol/liter	30	60	80	105